

Estimating the Number of Components in Panel Data Finite Mixture Regression Models with an Application to Production Function Heterogeneity

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Abstract

This paper develops likelihood-based methods for selecting the number of components in panel data finite mixture regression models. The panel structure ($T \geq 2$) eliminates the higher-order degeneracy of cross-sectional mixtures; we derive the asymptotic null distribution of the likelihood ratio test (LRT), propose a sequential bootstrap testing procedure for consistent model selection, and establish BIC consistency. We apply these methods to Chilean manufacturing plants, investigating production function heterogeneity under Hicks-neutral technology. Under the Nonlinear Mixture specification ($K_e=2$, $T=10$), we select $\hat{M}=2-6$ latent technology types per industry. Per-type GNR estimates reveal that the $\hat{\alpha}$ -weighted aggregate materials elasticity matches the pooled $M=1$ estimate, while the per-type distribution spans 0.44–0.74—a within-industry range of heterogeneity invisible to single-technology estimators. The data reject the single-technology hypothesis across all three specifications.

Keywords: finite mixture models, panel data, likelihood ratio test, Bayesian Information Criterion, production function heterogeneity, technology types

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1 Introduction

Finite mixture models provide a flexible and natural framework for representing unobserved heterogeneity across a finite number of latent classes. Due to their adaptability, these models have found extensive applications in empirical studies across various disciplines since the seminal introduction of a two-component normal mixture model by Pearson (1894). In economics, often utilizing panel data, finite mixture models have been particularly influential in capturing unobserved individual-specific effects in fields such as labour economics, health economics, and industrial organization.¹ The theoretical underpinnings and practical applications of finite mixture models have been extensively explored by Titterton et al. (1985), Lindsay (1995), and McLachlan and Peel (2004).

A critical aspect in the application of finite mixture models is determining the appropriate number of components, which often corresponds to the number of latent individual types or abilities in economic models. Selecting an incorrect number of components can lead to estimation biases, excessive computational costs, or identification difficulties. Therefore, developing reliable statistical procedures for accurately selecting the number of components in finite mixture models is essential for empirical analysis.

This paper investigates the statistical properties and practical performance of methods for determining the number of components in *panel data finite mixture regression models*, including the likelihood ratio test (LRT) and the Akaike and Bayesian Information Criteria (AIC and BIC). Panel data finite mixture models offer significant advantages over their cross-sectional counterparts by utilizing repeated observations for each unit, thereby enhancing parameter identification and mitigating common identification issues associated with cross-sectional analyses. However, the existing literature on selecting the number of components in panel data finite mixture regression models is limited, particularly when the component-specific regression errors are normally distributed—a commonly assumed baseline model in regression analyses. This study addresses this gap by rigorously examining and comparing these estimation methods.

Specifically, this paper develops the LRT and BIC methods for consistently selecting the number of components in panel data finite mixture regression models and applies these methods to examine production function heterogeneity. We first consider a baseline

¹For instance, Heckman and Singer (1984) utilize finite mixture models as an alternative method for handling unobserved heterogeneity in unemployment duration analysis. Similarly, Keane and Wolpin (1997) and Cameron and Heckman (1998) employ these models to analyze dynamic decision-making regarding schooling and occupational choices in the presence of unobserved heterogeneity in human capital. In health economics, Deb and Trivedi (1997) propose a finite mixture negative binomial model to capture the unobserved dispersion in healthcare utilization among elderly populations. Additionally, finite mixture models are extensively used in consumer segmentation within industrial organization, as demonstrated by Kamakura and Russell (1989) and Andrews and Currim (2003). Kasahara and Shimotsu (2009) and Hu and Shum (2012) establish the conditions under which finite mixture models are non-parametrically identified using panel data.

scenario where regression errors are conditionally independent and normally distributed across periods, conditional on the latent type. Acknowledging the limitations of the normality assumption, we also analyze cases where regression errors follow more flexible, component-specific normal mixtures. Our theoretical results in the main text concern the static (conditional-independence) panel mixture model. We also consider a nonparametric procedure to estimate a lower bound on the number of components for finite mixture models under conditional independence using the rank test of Kleibergen and Paap (2006) and Kasahara and Shimotsu (2014).

Testing for the number of components in normal mixture regression models remains challenging, as standard asymptotic regularity conditions fail due to parameter non-identifiability, singular Fisher information, and boundary parameter issues. While numerous studies have addressed the likelihood ratio test (LRT) for finite mixture models (e.g., Ghosh and Sen, 1985; Chernoff and Lander, 1995; Lemdani and Pons, 1997; Chen and Chen, 2001, 2003; Chen et al., 2004; Garel, 2001, 2005; Chen et al., 2014), deriving its asymptotic distribution through Gaussian processes (Dacunha-Castelle and Gassiat, 1999; Liu and Shao, 2003; Zhu and Zhang, 2004; Azaïs et al., 2009), these approaches fail in *cross-sectional* normal regression models due to: (i) infinite Fisher information for testing, (ii) unbounded log-likelihood function, and (iii) linear dependence between density derivatives for mean and variance parameters.²

To the best of our knowledge, whether the aforementioned problems (i)–(iii) of the cross-sectional normal mixture still arise in panel data normal finite mixture models or their extensions remains unknown in the literature. Furthermore, no likelihood-based test has yet been developed for testing the null hypothesis of an M_0 -component model against an alternative $(M_0 + 1)$ -component model for $M_0 \geq 1$ in panel normal regression mixture models with conditionally independent errors.

We show that the higher-order degeneracy of problem (iii) disappears in panel normal mixture models with conditional independence. The problems related to (i) and (ii) persist. Consequently, the existing approaches in Dacunha-Castelle and Gassiat (1999), Liu and Shao (2003), Zhu and Zhang (2004), and Azaïs et al. (2009) do not directly apply to this class of panel normal mixture models. We impose bounds on the component-specific variance parameters and mixing proportions to address the unboundedness and infinite Fisher information, respectively, and then analyze the asymptotic distribution of the LRT using reparameterization orthogonal to the direction in which the Fisher information matrix is singular. The likelihood ratio of an $(M_0 + 1)$ -component model against the M_0 -component model is approximated with local quadratic-form expansion with squares and cross-products of the reparameterized parameters. We demonstrate that the asymptotic

²Chen and Li (2009), Chen et al. (2012), and Kasahara and Shimotsu (2015) analyze the LRT asymptotics for univariate models, with Kasahara and Shimotsu (2019) extending to multivariate cases. Additionally, Amengual et al. (2022) develop a score-type test for cross-sectional normal mixtures.

null distributions of the LRT statistic are characterized by the maximum of M_0 random variables. Building on the LRT tests, we propose a sequential hypothesis testing approach for consistently estimating the number of components.

This paper contributes to the literature in several ways. First, we analyze the likelihood ratio test (LRT) for determining the number of components in panel data finite mixture regression models, where the regression errors follow component-specific normal mixtures. While Kasahara and Shimotsu (2015) and Kasahara and Shimotsu (2019) study the LRT for cross-sectional univariate and multivariate normal mixture regression models, respectively, we show that the asymptotic distribution of the LRT statistic in panel data differs due to the absence of higher-order dependencies when repeated outcome measurements are available under conditional independence.³

Second, while it is known that the log-likelihood function of normal mixture models is unbounded (Hartigan, 1985), whether this issue occurs in panel data remains unexplored. We show that the likelihood ratio test (LRT) statistic is also unbounded in panel normal mixture models with conditionally independent errors when the time dimension is finite, though this issue diminishes as the time dimension grows. This unboundedness may lead to excessive rejection rates in the LRT, which we address by imposing explicit bounds on the component-specific variance parameters.

Third, we establish the consistency of the Bayesian Information Criterion (BIC) and the inconsistency of the Akaike Information Criterion (AIC) for selecting the true number of components in panel data finite mixture models. Our consistency result generalizes Keribin (2000) by relaxing its restrictive condition (P2) through higher-order rank conditions and utilizing a generalized form of Le Cam’s differentiability in quadratic mean (DQM) framework (Liu and Shao, 2003; Kasahara and Shimotsu, 2018).

Fourth, we empirically analyze production technology heterogeneity using panel data from Chilean manufacturing plants in a two-stage procedure. In the first stage we select the number of latent technology types $\hat{M}$ using the sequential bootstrap LRT and BIC under three specifications — Cobb-Douglas, GNR Translog, and the Nonlinear Mixture ($K_\epsilon=2$) — selecting $\hat{M}=2-6$ per industry at $T=10$. In the second stage, conditional on $\hat{M}$, we recover type-specific output elasticities of capital, labour, and materials non-parametrically using the two-step procedure of Gandhi et al. (2020), with monotonicity projections that enforce the regularity conditions of production theory. Our findings re-

³From a technical perspective, our panel data finite mixture models with normally distributed errors constitute a special case of the multivariate normal framework studied by Kasahara and Shimotsu (2019). The primary distinction lies in our assumption of conditional independence within each mixture component, which facilitates improved identification of model parameters. We show that this structure yields asymptotic null distributions for the likelihood ratio test that differ from those in Kasahara and Shimotsu (2019), mainly because our panel data setting does not involve higher-order singularities. Furthermore, our extension to models with component-specific normal mixture errors is related to, but distinct from, standard multivariate normal models.

veal significant within-industry variation in output elasticities of material inputs and in TFP persistence, while constant-returns-to-scale holds within each type. This contrasts sharply with standard production function estimation methods, which typically impose homogeneous coefficients across plants (Olley and Pakes, 1996a; Levinsohn and Petrin, 2003a; Akerberg et al., 2015). Our results highlight the necessity of explicitly accounting for unobserved plant heterogeneity beyond Hicks-neutral technological differences in empirical analyses of production functions (Li and Sasaki, 2017; Doraszelski and Jaumandreu, 2018; Balat et al., 2019; Kasahara et al., 2022).

The identification and estimation of latent group structures in panel data has received attention in recent studies (Kasahara and Shimotsu, 2009; Lin and Ng, 2012; Bonhomme and Manresa, 2015; Ando and Bai, 2016; Su et al., 2016; Lu and Su, 2017). Finite mixture modeling provides a practical, model-based approach to determining unobserved group structures. Choosing the number of groups is often a prerequisite for classifying each individual's group membership. We can estimate the number of groups in panel data regression models by applying our proposed sequential hypothesis testing approach.

This paper closely relates to Kasahara et al. (2022), which analyzes the nonparametric identification and estimation of finite mixture production models with unobserved heterogeneity, explicitly assuming a known number of mixture components. In contrast, our study emphasizes testing and estimating the number of technology types, focusing specifically on heterogeneity in output elasticities of flexible inputs. By leveraging first-order condition expressions, our approach enables more flexible finite mixture model specifications than those studied by Kasahara et al. (2022). However, as our analysis does not recover the entire production function, our framework cannot address heterogeneity in output elasticities of predetermined inputs such as capital, nor heterogeneity in the stochastic processes governing Hicks-neutral technological changes. Thus, our paper complements Kasahara et al. (2022) by addressing distinct yet related aspects of mixture modeling in production function analysis.

The rest of this paper is organized as follows. In Section 2, we introduce several classes of panel data finite mixture regression models studied in this paper, along with empirical examples. In Section 4.2, we discuss the failure of regularity conditions when analyzing panel data finite mixture normal regression models. Section 4.3 analyzes the consistency of Maximum Likelihood Estimation (MLE). Section 5.1 analyzes the likelihood ratio test (LRT) for testing $H_0 : M_0 = 1$ against $H_1 : M_0 = 2$, while Section 5.2 considers the LRT for testing $H_0 : M_0 = M$ against $H_1 : M_0 = M + 1$ for $M \geq 2$. Section 5.3 develops a sequential hypothesis testing procedure. Section 5.4 discusses the estimation of a lower bound for the number of components using the rank test. Section 6 presents simulation results, while Section 7 presents empirical analysis.

In what follows, all limits are taken as $n \rightarrow \infty$ unless otherwise stated. Let $:=$ denote "equals by definition." For a $k \times 1$ vector $\mathbf{a}$ and a function $f(\mathbf{a})$, let $\nabla_{\mathbf{a}} f(\mathbf{a})$ denote the

$k \times 1$ vector of partial derivatives $(\partial/\partial \mathbf{a})f(\mathbf{a})$, and let $\nabla_{\mathbf{a}\mathbf{a}^\top} f(\mathbf{a})$ denote the $k \times k$ matrix of second partial derivatives $(\partial/\partial \mathbf{a}\partial \mathbf{a}^\top)f(\mathbf{a})$. Let $\|\cdot\|$ denote the Euclidean norm. We adopt the convention that capitalized letters, such as $\mathbf{W}$, represent random variables, whereas their lowercase counterparts, such as w , denote evaluation points. Let $\phi(t) := (2\pi)^{-1/2} \exp(-t^2/2)$ denote the standard normal density.

2 Heteroskedastic Finite Mixture Panel Regression Model

We consider finite mixture regression models with panel data, where the panel length $T \geq 2$ is fixed, and the number of cross-sectional observations n tends to infinity. Let M_0 denote the true number of components in the data-generating process, while M represents a candidate number of components in our model specification. Throughout this paper, we use asterisks to denote true parameter values; for instance, $\boldsymbol{\vartheta}_{M_0}^*$ represents the true parameter vector.

For each unit i , we specify the conditional probability density function of $\{y_{it}\}_{t=1}^T$ given $\{\mathbf{x}_{it}\}_{t=1}^T$ as:

$$g_M(\{y_{it}\}_{t=1}^T \mid \{\mathbf{x}_{it}\}_{t=1}^T; \boldsymbol{\vartheta}_M) = \sum_{j=1}^M \alpha_j f_j(\{y_{it}\}_{t=1}^T \mid \{\mathbf{x}_{it}\}_{t=1}^T; \boldsymbol{\theta}_j), \quad (1)$$

where $y_{it} \in \mathbb{R}$ and $\mathbf{x}_{it} \in \mathbb{R}^q$; α_j represents the population proportion of the j -th component, and $f_j(\{y_{it}\}_{t=1}^T \mid \{\mathbf{x}_{it}\}_{t=1}^T; \boldsymbol{\theta}_j)$ is the component-specific conditional density function with parameter vector $\boldsymbol{\theta}_j$. We collect the parameters into a vector as $\boldsymbol{\vartheta}_M := (\boldsymbol{\alpha}^\top, \boldsymbol{\theta}_1^\top, \dots, \boldsymbol{\theta}_M^\top)^\top \in \Theta_{\boldsymbol{\vartheta}_M}$, with $\boldsymbol{\alpha}^\top := (\alpha_1, \dots, \alpha_{M-1})^\top$ and $\alpha_M = 1 - \sum_{j=1}^{M-1} \alpha_j$. The q -dimensional vector $\mathbf{x}_{it}$ may include both time-varying and time-invariant regressors, and can accommodate correlated random effects specifications (Mundlak, 1978; Chamberlain, 1984) by including individual-specific averages or the complete sequence $\{\mathbf{x}_{it}\}_{t=1}^T$; see Appendix B.4.1 for a detailed treatment.

The component-specific density $f_j(\{y_{it}\}_{t=1}^T \mid \{\mathbf{x}_{it}\}_{t=1}^T; \boldsymbol{\theta}_j)$ in (1) assumes that outcomes $\{y_{it}\}_{t=1}^T$ are conditionally independent over time given covariates and component membership as:

$$f_j(\{y_{it}\}_{t=1}^T \mid \{\mathbf{x}_{it}\}_{t=1}^T; \boldsymbol{\theta}_j) = \prod_{t=1}^T f_{jt}(y_{it} \mid \mathbf{x}_{it}; \boldsymbol{\theta}_j), \quad (2)$$

where the period-specific density takes the form:

$$f_{jt}(y_{it} \mid \mathbf{x}_{it}; \boldsymbol{\theta}_j) = \sum_{k=1}^{K_\epsilon} \tau_{jk} \frac{1}{\sigma_j} \phi\left(\frac{y_{it} - \mu_{jk} - \mathbf{x}_{it}^\top \boldsymbol{\beta}_j}{\sigma_j}\right). \quad (3)$$

When $K_\epsilon = 1$, the density in (3) is normal with $\theta_j := (\mu_j, \sigma_j^2, \beta_j^\top)^\top \in \Theta_\theta$, where $\sigma_j^2 \in \mathbb{R}_{++}$ and $\beta_j \in \Theta_\beta$. When $K_\epsilon \geq 2$, it is a finite mixture of normals with $\theta_j := (\tau_j, \mu_j, \sigma_j^2, \beta_j^\top)^\top \in \Theta_\theta$, where $\tau_j = (\tau_{j1}, \dots, \tau_{j, K_\epsilon-1})^\top$ with $\tau_{jK_\epsilon} := 1 - \sum_{k=1}^{K_\epsilon-1} \tau_{jk}$, and $\mu_j = (\mu_{j1}, \dots, \mu_{jK_\epsilon})^\top$ satisfies $\mu_{j1} < \dots < \mu_{jK_\epsilon}$.

To simplify our analysis, we assume that the number of sub-components $K_\epsilon \geq 1$ within each mixture component is known to the researcher and that all true mixing proportions are strictly positive.⁴

Assumption 1. K_ϵ is known to the researcher. In case of $K_\epsilon \geq 2$, $\tau_{jk}^* > 0$, $\mu_{jk} \neq \mu_{j\ell}$, and $\mu_{1,jk} \neq \mu_{1,j\ell}$ if $k \neq \ell$ for $j = 1, \dots, M$ and $k, \ell = 1, \dots, K_\epsilon$.

The true number of components, M_0 , is defined as the smallest integer M such that the density admits a representation of the form (1).

3 Production Function Heterogeneity

We use Chilean manufacturing data to ask: *how many distinct production technologies are required to explain observed dispersion in material revenue shares?* By utilizing the first-order conditions with respect to flexible inputs, we investigate the presence of production function unobserved heterogeneity beyond Hicks-neutral productivity—specifically, the unobserved heterogeneity in output elasticities of flexible material input. Rather than fixing the number of technology types *a priori*, we treat it as an unknown parameter and estimate it with the finite-mixture framework of Section 2.

3.1 Data and common setup

Panel data structure. We observe a panel of N plants over T years. For plant i at time t the observable vector is $(Y_{it}, V_{it}, L_{it}, K_{it}, Z_{it})$, where Y_{it} is gross output, V_{it} intermediate materials, L_{it} labour, K_{it} capital, and Z_{it} a vector of additional plant or industry characteristics. Lower-case variables denote natural logs: $y_{it} = \ln Y_{it}$, $v_{it} = \ln V_{it}$, etc. Let P_t^Y be the output price, P_t^V the materials price index, and P_t^L the wage rate.

Timing and decision structure. Capital K_{it} is predetermined at the start of period t . The Hicks-neutral productivity shock ω_{it} is realised before the firm chooses flexible inputs. Following Gandhi et al. (2020), material is chosen after ω_{it} is observed but before an *ex post* shock to output ϵ_{it} is observed.

We formalise this timing protocol as follows.

⁴Testing the number of components K_ϵ simultaneously with the number of components M is certainly an important issue but beyond the scope of this paper, and left for future research.

Assumption 2 (Timing and Scalar Unobservable). (i) *Scalar unobservable*: The Hicks-neutral productivity ω_{it} is a scalar, realised and observed by the firm at the beginning of period t , before any variable input decisions are made.

(ii) *Predetermined capital, labour, and other plant characteristics*: L_{it} , K_{it} , and Z_{it} are chosen at $t - 1$ and is not adjustable within period t .

(iii) *Flexible inputs and unanticipated shocks*: Material V_{it} is chosen after ω_{it} is observed, conditional on $(\omega_{it}, L_{it}, K_{it}, Z_{it}, P_t^V, P_t^L, P_t^Y)$. An i.i.d. unanticipated ex post shock ϵ_{it} is then realised, independent of the firm's full information set: $\epsilon_{it} \perp (\omega_{it}, V_{it}, L_{it}, K_{it}, Z_{it}, P_t^V, P_t^L, P_t^Y)$. The distribution of ϵ_{it} may be different across latent types.

Assumption 2 follows the standard timing protocol in the production function literature (Olley and Pakes, 1996b; Levinsohn and Petrin, 2003b; Gandhi et al., 2020). Under Assumption 2(ii)-(iii), ℓ_{it} , k_{it} , and z_{it} are determined at or before $t - 1$. Instead of Assumption 2(ii)-(iii), we can alternatively assume that labour L_{it} is also a flexible input while adopting the differential timing assumption between L_{it} and V_{it} such that L_{it} is chosen before V_{it} as in Akerberg et al. (2015).

Latent Technology Types. Let $D_i \in \{1, 2, \dots, M\}$ denote the latent technology type of plant i , which determines the production function parameters and shock processes. Define the regressor vector $\mathbf{x}_{it} = (k_{it}, \ell_{it}, z_{it})^\top$.

3.2 Production function specification with Hicks-neutral technology

We consider three production function specifications: Cobb-Douglas, Translog (Lerner approximation), and Nonlinear Mixture—the exact first-order condition derived from the Translog production function, estimated with $K_e = 2$ mixture errors.

3.2.1 Cobb-Douglas

Under Cobb-Douglas technology,

$$y_{it} = \beta_{0,j} + \beta_{K,j}k_{it} + \beta_{L,j}\ell_{it} + \beta_{V,j}v_{it} + \omega_{it} + \epsilon_{it}. \quad (4)$$

The first-order condition for profit-maximising material choice yields⁵

$$s_{it}^V := \ln \left(\frac{P_t^V V_{it}}{P_t^Y Y_{it}} \right) = \ln \beta_{V,j} + \ln \mathbb{E}[e^{\epsilon_{it}} | D_i = j] - \epsilon_{it} =: c_{V,j} + u_{it}, \quad (5)$$

⁵Following Gandhi et al. (2020), this specification is obtained from a static expected profit maximisation

where $c_{V,j} := \ln \beta_{V,j} + \ln \mathbb{E}[e^{\epsilon_{it}} | D_i = j]$ is a type-specific constant and $u_{it} := -\epsilon_{it}$. Here, ϵ_{it} (and hence u_{it}) is serially independent over time under Assumption 2(iii).

Under a homogeneous production technology with $M = 1$, material output elasticities are common across plants, implying that material revenue shares should be constant up to an i.i.d. ex-post shock, as in (5). The observed dispersion in Figure 1 and Table 2 thus provides prima facie evidence against this homogeneity hypothesis. To allow for heterogeneous technologies, we consider the finite mixture model:

$$s_{it}^V = c_{V,j} + u_{it}, \quad j \sim \text{Discrete}(\alpha_1, \dots, \alpha_M), \quad (6)$$

where u_{it} follows either a normal or normal mixture distribution, and the number of types M is selected using the methods in Sections 5.1–5.3. The Cobb-Douglas restriction that material shares are constant within each type (up to the industry shifters $z'_{it}\gamma$) is testable. A likelihood ratio test against the Translog alternative (Section 3.2.2) strongly rejects this restriction in our empirical application (LRT statistic $p < 10^{-300}$; see Section 7). Consequently, the Cobb-Douglas results serve as a benchmark, while the Translog (Lerner approximation) and the Nonlinear Mixture specification remain our preferred specifications.

3.2.2 Translog approximation

The Translog specification allows output elasticities to vary with input levels:

$$\begin{aligned} y_{it} = & \beta_{0,j} + \beta_{K,j}k_{it} + \beta_{L,j}\ell_{it} + \beta_{V,j}v_{it} + \beta_{KK,j}k_{it}^2 + \beta_{LL,j}\ell_{it}^2 + \beta_{VV,j}v_{it}^2 \\ & + \beta_{KL,j}k_{it}\ell_{it} + \beta_{KV,j}k_{it}v_{it} + \beta_{LV,j}\ell_{it}v_{it} + \omega_{it} + \epsilon_{it}. \end{aligned} \quad (7)$$

The first-order condition yields

$$s_{it}^V = \ln(\beta_{V,j} + \beta_{VK,j}k_{it} + \beta_{VL,j}\ell_{it} + \beta_{VV,j}v_{it}) + \ln \mathbb{E}[e^{\epsilon_{it}} | D_i = j] - \epsilon_{it}. \quad (8)$$

Estimating the exact first-order condition in (8) directly within a finite mixture framework presents severe computational challenges. Specifically, the argument inside the logarithm must be strictly positive for all observations across all latent types j during numerical optimization. Imposing this global positivity constraint in a mixture model is notoriously difficult, making the resulting coefficient estimates highly sensitive to the maximum and

problem:

$$\max_{V_{it} \geq 0} \Pi_{it}(V_{it}; j) = \max_{V_{it} \geq 0} \left\{ P_t^Y B_{it,j} V_{it}^{\beta_{V,j}} \mathbb{E}[e^{\epsilon_{it}} | D_i = j] - P_t^V V_{it} \right\}$$

where $B_{it,j} := \exp(\beta_{0,j}) K_{it}^{\beta_{K,j}} L_{it}^{\beta_{L,j}} \exp(\omega_{it})$.

minimum realized values of the inputs k_{it} , ℓ_{it} , and v_{it} .

To circumvent these numerical instabilities, we consider an empirical specification—the *Lerner approximation*—based on the first-order Taylor expansion of (8) around the sample means $(\bar{k}, \bar{\ell}, \bar{v})$:

$$s_{it}^V = c_{V,j} + \beta_{VK,j}^* \tilde{k}_{it} + \beta_{VL,j}^* \tilde{\ell}_{it} + \beta_{VV,j}^* \tilde{v}_{it} + u_{it}, \quad (9)$$

where $\beta_{VX,j}^* := \beta_{VX,j} / \bar{\beta}_{V,j}$ with $\bar{\beta}_{V,j} := \beta_{V,j} + \beta_{VK,j} \bar{k} + \beta_{VL,j} \bar{\ell} + \beta_{VV,j} \bar{v}$ for $X \in \{K, L, V\}$, $\tilde{x}_{it} := x_{it} - \bar{x}$ are sample-demeaned inputs, and $c_{V,j} := \ln \bar{\beta}_{V,j} + \ln \mathbb{E}[e^{\epsilon_{it}} | D_i = j]$ is the log material elasticity evaluated at the sample means. Here, the error term u_{it} captures both the ex-post shock $-\epsilon_{it}$ and the linearization approximation error.

This specification is more flexible than the Cobb-Douglas baseline, allowing us to test whether technology types differ not only in average material intensity but also in how material shares respond to variations in capital, labour, and material inputs. For example, the coefficient $\beta_{VK,j}^*$ is the normalized interaction parameter $\beta_{VK,j} / \bar{\beta}_{V,j}$ from the Translog production function (7).⁶ It measures how the output elasticity of materials varies with capital intensity rather than reflecting the direct capital-output elasticity. A negative $\beta_{VK,j}^*$ indicates that capital-intensive plants use materials less intensively at the margin, which is consistent with capital–material substitutability.

Two scoping limitations of this approach are worth noting. First, computing the full set of Translog parameters requires estimating the remaining production function coefficients ($\beta_{K,j}$, $\beta_{KK,j}$, and $\beta_{KL,j}$) via the second step of the Gandhi et al. (2020) non-parametric framework under latent heterogeneity. This sequential estimation is beyond the scope of the current paper; see Kasahara et al. (2022) for a related implementation. Second, the linearization residual is potentially heteroskedastic and correlated with the regressors due to an approximation error. We address this approximation concern in the empirical application through the *Nonlinear Mixture* specification (Section 7), which estimates the exact Translog condition (8) directly with $K_\epsilon = 2$ mixture errors, bypassing the Lerner approximation.

⁶By Young’s theorem on the symmetry of second-order partial derivatives, $\beta_{VK,j} = \beta_{KV,j}$.

4 Maximum Likelihood Estimation

4.1 Data-Generating Process

We consider a random sample of n observations $\{\mathbf{W}_i\}_{i=1}^n$, where $\mathbf{W}_i = \{(Y_{it}, \mathbf{X}_{it}^\top)^\top\}_{t=1}^T$ is drawn from an M_0 -component density $g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)$ defined in equation (10):

$$g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*) = \sum_{j=1}^{M_0} \alpha_j^* f_j(\mathbf{w}; \boldsymbol{\theta}_j^*), \quad (10)$$

where $\boldsymbol{\vartheta}_{M_0}^* = (\boldsymbol{\theta}_1^{*\top}, \boldsymbol{\theta}_2^{*\top}, \dots, \boldsymbol{\theta}_{M_0}^{*\top}, \alpha_1^*, \dots, \alpha_{M_0-1}^*)^\top \in \Theta_{\boldsymbol{\vartheta}_{M_0}}$ and $\alpha_{M_0}^* = 1 - \sum_{j=1}^{M_0-1} \alpha_j^*$. For identification, we assume $\mu_1^* < \mu_2^* < \dots < \mu_{M_0}^*$ in the true parameters.

In Section 3, M_0 represents the number of latent technology types, reflecting plant-level heterogeneity in output elasticities. Our objective is to test whether $M_0 = 1$ (no heterogeneity) or $M_0 \geq 2$ (heterogeneous types exist).

4.2 The Testing Problem and Regularity Failures

To examine the failure of the regularity conditions of the likelihood ratio test statistic (LRTS) in panel finite mixture models, consider testing the null hypothesis $H_0 : M_0 = 1$ against the alternative hypothesis $H_1 : M_0 = 2$, where $\mathbf{W}_i = \{(Y_{it}, \mathbf{X}_{it}^\top)^\top\}_{t=1}^T$ is drawn from a true one-component density $f(\mathbf{w}; \boldsymbol{\theta}^*) = \prod_{t=1}^T f(y_t | \mathbf{x}_t; \boldsymbol{\theta}^*)$.

The two-component model

$$g_2(\mathbf{w}; \boldsymbol{\vartheta}_2) = \alpha f(\mathbf{w}; \boldsymbol{\theta}_1) + (1 - \alpha) f(\mathbf{w}; \boldsymbol{\theta}_2) \quad (11)$$

can generate the true one-component density in two cases: (1) $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2 = \boldsymbol{\theta}^*$ with $\alpha \in (0, 1)$, and (2) $\alpha \in \{0, 1\}$ with arbitrary $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$. Consequently, $H_0 : M_0 = 1$ can be partitioned into two sub-hypotheses: $H_{01} : \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$ and $H_{02} : \alpha(1 - \alpha) = 0$. The regularity conditions of the LRTS fail under both sub-hypotheses: under H_{01} , α is not identified and the Fisher information matrix for the other parameters becomes singular; under H_{02} , α is on the boundary of the parameter space and either $\boldsymbol{\theta}_1$ or $\boldsymbol{\theta}_2$ is not identified.

Three Fundamental Challenges in Normal Mixtures Analyzing the asymptotic distribution of the LRTS for *cross-sectional* normal mixtures (i.e., with $T = 1$) is notoriously challenging due to three fundamental problems (cf., Chen and Li, 2009):

- (i) **Infinite Fisher Information:** The Fisher information for testing H_{02} is not finite.
- (ii) **Unbounded Likelihood:** The log-likelihood function is unbounded (Hartigan, 1985).

(iii) **Linear Dependence of Derivatives:** The first-order derivative of $g_2(\mathbf{w}; \boldsymbol{\vartheta}_2)$ with respect to σ_j^2 is linearly dependent on its second-order derivative with respect to μ_j .

A central question is whether problems (i)–(iii) persist in panel normal mixture models with $T \geq 2$. We now examine each problem systematically.

Problem (i): Infinite Fisher Information Persists The issue of infinite Fisher information for testing $H_{02} : \alpha = 0$ arises in panel normal mixture models. The score for testing $H_{02} : \alpha = 0$ takes the form

$$\left. \frac{\partial \log g_2(\mathbf{W}; \alpha, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\partial \alpha} \right|_{\alpha=0, \boldsymbol{\theta}_2=\boldsymbol{\theta}^*} = \frac{f(\mathbf{W}; \boldsymbol{\theta}_1)}{f(\mathbf{W}; \boldsymbol{\theta}^*)} - 1, \quad (12)$$

where $f(\mathbf{W}; \boldsymbol{\theta}) = \prod_{t=1}^T \phi((Y_t - \mu - \mathbf{X}_t^\top \boldsymbol{\beta})/\sigma)/\sigma$. Then, $\mathbb{E}[\{f(\mathbf{W}; \boldsymbol{\theta}_1)/f(\mathbf{W}; \boldsymbol{\theta}^*) - 1\}^2] = \infty$ when $\sigma_1^2 > 2\sigma^{*2}$. This infinite Fisher information also arises for normal mixture density (2)–(3). For more details, see Proposition 5.

Implication: We focus on testing H_{01} by restricting $\alpha \in [c_1, 1 - c_1]$ for some $c_1 > 0$, sidestepping the infinite Fisher information under H_{02} . Appendix A.1 treats H_{02} under $\sigma_1^2 \leq 2\sigma^{*2} - c$ for some $c > 0$ following Andrews (2001).

Problem (ii): Unbounded Likelihood Persists The LRTS in normal mixture models with panel data becomes unbounded as the sample size n goes to ∞ . Define the likelihood ratio statistic with respect to the true parameter under H_0 as

$$LR_n^*(\boldsymbol{\vartheta}_2) := 2 \left\{ \sum_{i=1}^n \log g_2(\mathbf{W}_i; \boldsymbol{\vartheta}_2) - \sum_{i=1}^n \log f(\mathbf{W}_i; \boldsymbol{\theta}^*) \right\}, \quad (13)$$

where g_2 is the density of the two-component finite mixture distribution in (1) with $M = 2$ and $\boldsymbol{\theta}^*$ is the true parameter value under H_0 .

Proposition 1. *Suppose that the true model is described by the one-component model with $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ and that the parameter space for σ_j^2 is $\Theta_{\sigma^2} = \mathbb{R}_{++}$. Then, for any positive constant $M > 0$, when the component-specific density function is given by (2) with (3),*

$$\Pr \left(\sup_{\boldsymbol{\vartheta}_2 \in \Theta_{\boldsymbol{\vartheta}_2}} LR_n^*(\boldsymbol{\vartheta}_2) < M \right) = \exp(-C_T n^{1/T}) \quad \text{as } n \rightarrow \infty, \quad (14)$$

where C_T for $T = 2, 3, \dots$ are positive constants depending on M and T .

Problem (iii): Linear Dependence Resolved for $T \geq 2$ Consider the two-component panel mixture model without covariates:

$$g_2(\{y_t\}_{t=1}^T; \alpha, \theta_1, \theta_2) = \alpha f(\{y_t\}_{t=1}^T; \theta_1) + (1 - \alpha) f(\{y_t\}_{t=1}^T; \theta_2) \quad \text{with}$$

$$f(\{y_t\}_{t=1}^T; \theta_j) = \prod_{t=1}^T \frac{1}{\sigma_j} \phi\left(\frac{y_t - \mu_j}{\sigma_j}\right) \quad \text{for } j = 1, 2, \quad (15)$$

where $\theta_j = (\mu_j, \sigma_j^2)^\top$. When evaluated under the null hypothesis $H_{01} : \theta_1 = \theta_2 = \theta^*$ with $\alpha(1 - \alpha) \neq 0$, the first-order derivative of $g_2(\{y_t\}_{t=1}^T; \alpha, \theta_1, \theta_2)$ with respect to σ_j^2 is *not* linearly dependent on its second-order derivative with respect to μ_j .

Proposition 2. For any $j \in \{1, 2\}$, any $a, b \in \mathbb{R}$, and any $\bar{\alpha} \in [c, 1 - c]$ for some constant $c > 0$, when $T \geq 2$:

$$\Pr\left(\frac{\partial^2 g_2(\{Y_t\}_{t=1}^T; \alpha, \theta_1, \theta_2)}{(\partial \mu_j)^2} \bigg|_{(\alpha, \theta_1, \theta_2) = (\bar{\alpha}, \theta^*, \theta^*)} = a + b \frac{\partial g_2(\{Y_t\}_{t=1}^T; \alpha, \theta_1, \theta_2)}{\partial \sigma_j^2} \bigg|_{(\alpha, \theta_1, \theta_2) = (\bar{\alpha}, \theta^*, \theta^*)}\right) = 0. \quad (16)$$

Implication: The panel mixture model (1) with (2)-(3) is *strongly identifiable* (Chen, 1995): the best convergence rate for the mixing distribution is $n^{-1/4}$ (vs. $n^{-1/8}$ for cross-sectional normal mixtures; Kasahara and Shimotsu, 2015). The key is the product structure of the component density across time periods under conditional independence.

The Remaining Challenge: Non-Identification of α Under H_{01} Despite the resolution of problem (iii), the standard asymptotic analysis breaks down in testing $H_{01} : \theta_1 = \theta_2 = \theta^*$ because α is not identified under H_{01} . In addition, the first-order derivatives at the true value $\vartheta_2^* = (\alpha, (\theta^*)^\top, (\theta^*)^\top)^\top$ are linearly dependent:

$$\nabla_{\theta_1} \log g_2(w; \vartheta_2^*) = \frac{\alpha}{1 - \alpha} \nabla_{\theta_2} \log g_2(w; \vartheta_2^*). \quad (17)$$

Consequently, we derive the asymptotic distribution of the LRTS using a fourth-order Taylor series approximation for the log-likelihood function.

4.3 Restricted MLE and Consistency

To address the issue of unboundedness, following Hathaway (1985), we restrict the parameter space to ensure that the ratio of each component-specific variance to the unconditional variance is bounded away from zero, i.e., $\min_j \left(\frac{\sigma_j^2}{\sum_{k=1}^M \alpha_k \sigma_k^2} \right) \geq c_2$, for some

constant $c_2 > 0$. This constraint enforces a positive lower bound on the component variances when the unconditional variance is strictly positive with $\sum_{k=1}^M \alpha_k \sigma_k^2 \geq c_3 > 0$. We also assume that α_j is bounded away from 0 and 1 by a small constant $c_1 > 0$ so that $\alpha \in \Theta_{\alpha, c_1} := \{\alpha : \alpha_j \in [c_1, 1 - c_1], \sum_{j=1}^{M-1} \alpha_j \leq 1 - c_1\}$. This assumption prevents issues related to infinite Fisher information, allowing us to focus on H_{01} . Let $\mathbf{c} = (c_1, c_2, c_3)^\top$, and define a restricted parameter space for ϑ_M for a model with the component-specific density function (2) with (3) by

$$\bar{\Theta}_{\vartheta_M}(\mathbf{c}) := \{\vartheta_M \in \Theta_{\vartheta_M} : \alpha \in \Theta_{\alpha, c_1}, \min_j \sigma_j^2 \geq c_2 \sum_{k=1}^M \alpha_k \sigma_k^2, \sum_{k=1}^M \alpha_k \sigma_k^2 \geq c_3\}.$$

We consider the MLE of ϑ_M over the restricted parameter space $\bar{\Theta}_{\vartheta_M}(\mathbf{c})$ for the M -component model:

$$\ell_n^M(\hat{\vartheta}_M) = \sup_{\vartheta_M \in \bar{\Theta}_{\vartheta_M}(\mathbf{c})} \ell_n^M(\vartheta_M), \quad (18)$$

where $\ell_n^M(\vartheta_M) = \sum_{i=1}^n \log g_M(\mathbf{W}_i; \vartheta_M)$ is the log-likelihood function. The MLE $\hat{\vartheta}_M$ may not be unique when M exceeds the true number of components.

For $M > M_0$, define the set of parameter values that yield the true M_0 -component density as

$$\Theta_{\vartheta_M}^* := \{\vartheta_M \in \Theta_{\vartheta_M} : g_M(\mathbf{w}; \vartheta_M) = g_{M_0}(\mathbf{w}; \vartheta_{M_0}^*) \text{ for all } \mathbf{w} \in \mathcal{W}\}.$$

For instance, if $M_0 = 1$ with true density $f(\mathbf{w}; \theta^*)$, then

$$\Theta_{\vartheta_2}^* = \{(\alpha, \theta_1, \theta_2) \in \Theta_{\vartheta_2} : (\alpha = 1, \theta_1 = \theta^*) \text{ or } (\alpha = 0, \theta_2 = \theta^*)\},$$

where θ_1 and θ_2 are component-specific parameters. The following proposition establishes the consistency of the MLE in over-specified mixture models.

Assumption 3. (a) $T \geq 2$ and $\mathbf{X}$ has finite second moments and $\Pr(\mathbf{X}^\top \beta_j \neq \mathbf{X}^\top \beta_j^*) > 0$ for $\beta_j \neq \beta_j^*$ for $j = 1, 2, \dots, M_0$. (b) For finite $M > M_0$, $\Theta_{\vartheta_M}^* \cap \bar{\Theta}_{\vartheta_M}(\mathbf{c})$ is non-empty. (c) $\bar{\Theta}_{\vartheta_M}(\mathbf{c})$ is compact.

Proposition 3. Suppose that Assumptions 1 and 3 hold and that the data are generated from M_0 -component models with the component-specific density function (2) with (3). Then, for any finite $M > M_0$, $\inf_{\vartheta_M \in \Theta_{\vartheta_M}^*} \|\hat{\vartheta}_M - \vartheta_M\| \rightarrow 0$ almost surely.

Consequently, Proposition 3 suggests that the MLE $\hat{\vartheta}_M$ converges almost surely to a set of parameters for which the true density function $g_{M_0}(\mathbf{w}; \theta_{M_0}^*)$ emerges within the space of overspecified M -component density functions.

5 Likelihood Ratio Test for the Number of Components

5.1 Likelihood ratio test for $H_0 : M_0 = 1$ against $H_1 : M_0 = 2$

We first consider testing $H_0 : M_0 = 1$ against $H_1 : M_0 = 2$. Let $\hat{\theta}_0$ be the one-component MLE and define the LRTS as

$$LR_n := 2 \{ \ell_n^2(\hat{\vartheta}_2) - \ell_n^1(\hat{\theta}_0) \}, \quad (19)$$

where $\hat{\vartheta}_2$ is the MLE under the restricted parameter space defined by (18).

Deriving the asymptotic distribution of the LRTS is non-standard because the Fisher information matrix is singular under the null hypothesis. To address this, we reparameterize θ_1 and θ_2 following Kasahara and Shimotsu (2012):

$$\begin{pmatrix} \lambda \\ \nu \end{pmatrix} := \begin{pmatrix} \theta_1 - \theta_2 \\ \alpha\theta_1 + (1 - \alpha)\theta_2 \end{pmatrix}, \quad (20)$$

where λ captures deviations from the one-component model such that $\lambda = \mathbf{0}$ under H_0 . Standard quadratic approximation fails because $\nabla_\lambda \log g_2(w; \alpha, \nu + (1 - \alpha)\lambda, \nu - \alpha\lambda)|_{\lambda=0} = \mathbf{0}$. We therefore use second-order derivatives with respect to λ for identification.

Define the score function $s(\mathbf{W}) = (s_\nu(\mathbf{W})^\top, s_{\lambda\lambda}(\mathbf{W})^\top)^\top$, where $s_\nu(\mathbf{W}) := \nabla_\theta f^*/f^*$ and $s_{\lambda\lambda}(\mathbf{W}) := \frac{1}{2} \widehat{\text{vech}_2}(\nabla_{\theta\theta^\top} f^*/f^*)$ with $f^* = f(\mathbf{W}; \theta^*)$. The function $s_{\lambda\lambda}(w)$ comprises second-order derivatives of the log-likelihood and serves as the score for identifying λ . Explicit expressions via Hermite polynomials are in Appendix A.3.2, which establishes non-singularity of $\mathcal{I} := \mathbb{E}[s(\mathbf{W})s(\mathbf{W})^\top]$.

Define the information matrix partition and related quantities:

$$\mathcal{I} = \begin{pmatrix} \mathcal{I}_\nu & \mathcal{I}_{\nu\lambda} \\ \mathcal{I}_{\lambda\nu} & \mathcal{I}_{\lambda\lambda} \end{pmatrix}, \quad \mathcal{I}_{\lambda,\nu} := \mathcal{I}_{\lambda\lambda} - \mathcal{I}_{\lambda\nu} \mathcal{I}_\nu^{-1} \mathcal{I}_{\nu\lambda}, \quad \mathcal{G}_{\lambda,\nu} := (\mathcal{I}_{\lambda,\nu})^{-1} \mathcal{S}_{\lambda,\nu}, \quad (21)$$

where $\mathcal{G}_{\lambda,\nu} \sim \mathcal{N}(\mathbf{0}, (\mathcal{I}_{\lambda,\nu})^{-1})$. Here $\mathcal{S}_{\lambda,\nu} := n^{-1/2} \sum_{i=1}^n s_{\lambda,i} - \mathcal{I}_{\lambda\nu} \mathcal{I}_\nu^{-1} n^{-1/2} \sum_{i=1}^n s_{\nu,i}$ is the score for λ projected orthogonal to ν .

Let $\hat{t}_\lambda$ be the projection of $\mathcal{G}_{\lambda,\nu}$ onto the cone $\Lambda_\lambda := \{v(\lambda) : \lambda \in \mathbb{R}^q\}$, where

$$v(\lambda) = (\lambda_1^2, \lambda_2 \lambda_1, \lambda_2^2, \dots, \lambda_q^2)^\top \quad (22)$$

contains unique elements of $\lambda\lambda^\top$. Formally,

$$\hat{t}_\lambda = \arg \min_{t_\lambda \in \Lambda_\lambda} (t_\lambda - \mathcal{G}_{\lambda,\nu})^\top \mathcal{I}_{\lambda,\nu} (t_\lambda - \mathcal{G}_{\lambda,\nu}). \quad (23)$$

Assumption 4. (a) $\mathbf{X}$ has finite 9th moments. (b) $\mathbb{E}[\mathbf{U}\mathbf{U}^\top]$ is non-singular, where $\mathbf{U} = [1, \mathbf{X}^\top]^\top$. (c) $\mathcal{I}$ is non-singular. (d) The true parameter $\boldsymbol{\theta}^*$ is interior to Θ .

Proposition 4. Suppose Assumptions 1-4 hold and data are generated under $H_0 : M_0 = 1$. Then, for models (1) with the component density function (2) with (3) $LR_n \xrightarrow{d} \widehat{\mathbf{t}}_\lambda^\top \mathcal{I}_{\lambda, \nu} \widehat{\mathbf{t}}_\lambda$.

The non-singularity of $\mathcal{I}$ for model (1) with (2)–(3) when $K_\epsilon = 1$ is proved in Lemma 4. For $K_\epsilon = 2$, non-singularity is established analytically in Lemma 6 under the panel-covariate joint identifying variation condition (Assumption MOD-1), which holds in our empirical specifications; Remark 2 extends the argument to $K_\epsilon \geq 3$ via iterated phase-peeling.

5.2 Likelihood ratio test for $H_0 : M_0 = M$ against $H_1 : M_0 = M + 1$

Testing $H_0 : M_0 = M$ in the $(M + 1)$ -component model requires addressing two scenarios: (i) two components have identical parameters, or (ii) one component has zero mixing proportion. We partition H_0 as $H_0 = H_{01} \cup H_{02}$, where $H_{01} : \boldsymbol{\theta}_h = \boldsymbol{\theta}_{h+1}$ for some $h \in \{1, \dots, M\}$ with $\alpha_j > 0$ for all j , and $H_{02} : \alpha_h = 0$ for some h .

Issue with testing H_{02} : When $\alpha_h = 0$, the component-specific parameter $\boldsymbol{\theta}_h$ is unidentified, leading to infinite Fisher information unless a priori restrictions are imposed on component variances.

Proposition 5. $\sup_{\boldsymbol{\vartheta}_{M_0+1} \in \Theta_{\boldsymbol{\vartheta}_{M_0+1, 2h}}^*} \mathbb{E}[\{\nabla_{\alpha_h} \log g_{M_0+1}(\mathbf{W}, \boldsymbol{\vartheta}_{M_0+1})\}^2] < \infty$ if and only if $\max\{\sigma^2 : \sigma \in \Theta_\sigma\} < 2 \max\{\sigma_1^{2*}, \dots, \sigma_{M_0}^{2*}\}$ for $g_{M_0+1}(\mathbf{w}, \boldsymbol{\vartheta}_{M_0+1})$ in (1) with the component-specific density function (2) with (3).

Because the variance restriction in Proposition 5 is difficult to justify and enforce, we focus on testing H_{01} .

Testing H_{01} : Partition $H_{01} = \cup_{h=1}^{M_0} H_{0,1h}$, where $H_{0,1h} : \boldsymbol{\theta}_h = \boldsymbol{\theta}_{h+1}$ with ordered means $\mu_1 < \dots < \mu_h = \mu_{h+1} < \dots < \mu_{M_0+1}$. Each h corresponds to one way of describing the M_0 -component null model in the $(M_0 + 1)$ -component space.

We construct local MLEs $\widehat{\boldsymbol{\vartheta}}_{M_0+1}^h$ by restricting the $(M_0 + 1)$ -component estimator to a neighborhood $\widehat{\Psi}_h^*$ that contains parameters satisfying $H_{0,1h}$ but excludes other $H_{0,1l}$ for $l \neq h$. The formal definition of Ψ_h^* and partition $\Theta_{\boldsymbol{\theta}, h}^*$ are in Appendix A.3. Define the local LRTS:

$$LR_n^{M_0, h} := 2 \left\{ \ell_n^{M_0+1}(\widehat{\boldsymbol{\vartheta}}_{M_0+1}^h) - \ell_n^{M_0}(\widehat{\boldsymbol{\vartheta}}_{M_0}) \right\} \quad \text{for } h = 1, \dots, M_0.$$

Because $\cup_{h=1}^{M_0} \Psi_h^* = \bar{\Theta}_{\vartheta_{M_0+1}}(c)$, the global LRTS is the maximum of local LRTS:

$$LR_n^{M_0} := 2 \left\{ \ell_n^{M_0+1}(\hat{\vartheta}_{M_0+1}) - \ell_n^{M_0}(\hat{\vartheta}_{M_0}) \right\} = \max\{LR_n^{M_0,1}, \dots, LR_n^{M_0,M_0}\}. \quad (24)$$

Each $LR_n^{M_0,h}$ converges to $(\hat{t}_\lambda^h)^\top \mathcal{I}_{\lambda,\nu}^h \hat{t}_\lambda^h$, analogous to the one-component case. Thus, the asymptotic distribution of $LR_n^{M_0}$ is the maximum of M_0 such random variables.

Assumption 5. (a) Assumption 4(a)(b) hold. (b) $\mathcal{I}$ defined in (47) is finite and non-singular. (c) The true parameter $\vartheta_{M_0}^*$ is interior to $\bar{\Theta}_{\vartheta_{M_0}}(c)$.

Proposition 6. Suppose Assumptions 1, 3, and 5 hold. Under $H_0 : M_0 = M$, for models (1) with component density function (2) with (3),

$$LR_n^{M_0} \xrightarrow{d} \max_{h=1, \dots, M_0} \{(\hat{t}_\lambda^h)^\top \mathcal{I}_{\lambda,\nu}^h \hat{t}_\lambda^h\},$$

where $(\hat{t}_\lambda^h)^\top \mathcal{I}_{\lambda,\nu}^h \hat{t}_\lambda^h$ is defined in (48).

Implementation: In practice, we use parametric bootstrap to obtain p-values for testing $H_0 : M_0 = M$ against $H_1 : M_0 = M + 1$ (see, e.g., Kasahara and Shimotsu, 2019). The parametric bootstrap is preferred because it circumvents the need to compute $\mathcal{I}_{\lambda,\nu}$ and to project onto the cone Λ_λ , both of which are computationally demanding when q is moderate.

Regularized estimation. The theoretical results above are derived for the restricted MLE over $\bar{\Theta}_{\vartheta_M}(c)$. In estimation, the EM algorithm maximises the penalised log-likelihood $\ell_n^{\text{pen}}(\vartheta) := \ell_n(\vartheta) + p(\vartheta)$ over $\bar{\Theta}_{\vartheta_M}(c)$, where $p(\vartheta)$ collects the log-prior kernels from the following regularisation devices:

- (i) **Variance floor:** $\sigma_j^2 \geq \delta \hat{s}^2$ with $\delta = 0.05$, where $\hat{s}^2 = \text{Var}(y)$ is the pooled sample variance.
- (ii) **Penalty on mixing proportions:** a logarithmic penalty $(a_\alpha - 1) \sum_{j=1}^M \log \alpha_j$ on $(\alpha_1, \dots, \alpha_M)$ with $a_\alpha = 3$, which discourages mixing proportions α_j from collapsing to zero.
- (iii) **Penalty on sub-component weights:** a logarithmic penalty $(a_\tau - 1) \sum_{k=1}^{K_\epsilon} \log \tau_{jk}$ on $(\tau_{j1}, \dots, \tau_{jK_\epsilon})$ with $a_\tau = 1.5$, applied per type.

Only the variance has a hard lower bound (item (i)); the mixing proportions α_j and the sub-component weights τ_{jk} are kept away from the simplex boundary by the logarithmic penalties in items (ii)–(iii) alone. The penalty derivatives diverge as $\alpha_j \rightarrow 0$ or $\tau_{jk} \rightarrow 0$, providing smooth boundary repulsion that obviates a separate hard bound while preserving asymptotic equivalence to the unconstrained MLE on the interior of the parameter space. The explicit penalty $p(\vartheta)$ is given in equation (76) (Remark 1). Since $p(\vartheta) = O(1)$ at

any interior point of the parameter space, the penalised MLE is first-order asymptotically equivalent to the restricted MLE; see the appendix for a formal argument.

The following proposition establishes that the parametric bootstrap is asymptotically valid.

Proposition 7 (Bootstrap validity of the parametric bootstrap LRT). *Suppose Assumptions 1 and 3 hold, the number of bootstrap replications satisfies $B_n \rightarrow \infty$ as $n \rightarrow \infty$, and the true number of components is M_0 .*

- (a) **Testing** $H_0: M_0 = 1$ *vs.* $H_1: M_0 = 2$. *Suppose Assumption 4 holds. Let F_1 denote the CDF of the limiting null distribution of LR_n in Proposition 4. Generate bootstrap samples $\mathbf{W}_1^*, \dots, \mathbf{W}_n^*$ i.i.d. from the estimated one-component density $g_1(\cdot; \hat{\boldsymbol{\theta}}_1)$, and let LR_n^* denote the LRT statistic computed from the bootstrap sample. Then, at every continuity point x of F_1 ,*

$$P^*(LR_n^* \leq x \mid \{\mathbf{W}_i\}_{i=1}^n) - F_1(x) \xrightarrow{p} 0.$$

- (b) **Testing** $H_0: M_0 = M$ *vs.* $H_1: M_0 = M + 1$ **for** $M \geq 2$. *Suppose additionally that Assumption 5 holds. Let F_M denote the CDF of the limiting null distribution of $LR_n^{M_0}$ in Proposition 6. Generate bootstrap samples $\mathbf{W}_1^*, \dots, \mathbf{W}_n^*$ i.i.d. from $g_M(\cdot; \hat{\boldsymbol{\theta}}_M)$. Then, at every continuity point x of F_M ,*

$$P^*(LR_n^{*,M} \leq x \mid \{\mathbf{W}_i\}_{i=1}^n) - F_M(x) \xrightarrow{p} 0.$$

- (c) **Consistency of bootstrap sequential testing.** *Under the conditions of Proposition 8, and with $B_n q_n \rightarrow \infty$ so that simulated bootstrap quantiles are consistent in the tail, the bootstrap sequential estimator $\hat{M}_{\text{LRT}}$ using bootstrap critical values satisfies $\hat{M}_{\text{LRT}} - M_0 = o_p(1)$.*

The proof in Appendix A.6 follows the local-null stability argument developed in this paper, combined with the bootstrap-continuity template in Lehmann and Romano (2005, Theorem 15.4.2) (see also Kasahara and Shimotsu (2019, Proposition 6 and Lemma 12)). Remark 1 in the appendix verifies that the penalised bootstrap used in implementation is first-order asymptotically equivalent.

Finite-sample validity is confirmed by the size tests in Appendix Table B5, which show that the bootstrap LRT maintains empirical rejection rates close to nominal levels across all regularisation settings used in our empirical application.

5.3 Sequential hypothesis testing

To estimate the number of components, we sequentially test $H_0: M_0 = M$ against $H_1: M_0 = M + 1$ starting from $M = 1$, and then $M = 2, \dots, \bar{M}$, where $\bar{M}$ is the upper bound for

the number of components, which is assumed to be known and larger than M_0 . The first value of M that leads to a nonrejection of H_0 gives our estimate for M_0 . Robin and Smith (2000) develop a similar sequential hypothesis test for estimating the rank of a matrix.

For $M = 1, \dots, \bar{M}$, let $c_{1-q_n}^M$ denote the $100(1 - q_n)$ percentile of the cumulative distribution function of a random variable $\max\{(\hat{t}_\lambda^1)^\top \mathcal{I}_{\lambda,\nu}^1 \hat{t}_\lambda^1, \dots, (\hat{t}_\lambda^M)^\top \mathcal{I}_{\lambda,\nu}^M \hat{t}_\lambda^M\}$ for testing $H_0 : M_0 = M$ in Proposition 6. Let $\hat{c}_{1-q_n}^M$ be a consistent estimator of $c_{1-q_n}^M$. Then, our estimator based on sequential hypothesis testing (SHT, hereafter) is defined as

$$\hat{M}_{\text{LRT}} = \min_{M \in \{1, \dots, \bar{M}\}} \{M : LR_n^M \geq \hat{c}_{1-q_n}^M \text{ for } M = 1, \dots, M-1, \text{ and } LR_n^M < \hat{c}_{1-q_n}^M\}. \quad (25)$$

The estimator $\hat{M}_{\text{LRT}}$ depends on the choice of the significance level q_n . The following proposition states that $\hat{M}_{\text{LRT}}$ converge to M_0 in probability as $n \rightarrow \infty$ when $-n^{-1} \log q_n = o(1)$ and $q_n = o(1)$.

Let

$$Q_n^M(\vartheta_M) := n^{-1} \sum_{i=1}^n \log g_M(\mathbf{W}_i; \vartheta_M) \text{ and } Q^M(\vartheta_M) := \mathbb{E}[\log g_M(\mathbf{W}_i; \vartheta_M)].$$

Assumption 6. For $M = 1, \dots, M_0$, (a) $Q^M(\vartheta_M)$ has a unique maximum at ϑ_M^* in Θ_{ϑ_M} ; (b) Θ_{ϑ_M} is compact; (c) ϑ_M^* is interior to Θ_{ϑ_M} ; (d) $B^M(\vartheta_M^*) := \mathbb{E}\left\{\nabla_{\vartheta_M} \log g_M(\mathbf{W}_i; \vartheta_M) \nabla_{\vartheta_M}^\top \log g_M(\mathbf{W}_i; \vartheta_M)\right\}$ is non-singular; (e) $A^M(\vartheta_M^*) := \mathbb{E}\left\{\nabla_{\vartheta_M \vartheta_M^\top} \log g_M(\mathbf{W}_i; \vartheta_M)\right\}$ has a constant rank in some open neighborhood of ϑ_M^* ; and (f) $Q^{M+1}(\vartheta_{M+1}^*) - Q^M(\vartheta_M^*) > 0$ for $M = 1, \dots, M_0 - 1$.

Proposition 8. Suppose that $M_0 < \bar{M}$ and Assumptions 1, 4, 5, and 6 hold. If we choose q_n such that $-n^{-1} \log q_n = o(1)$ and $q_n = o(1)$, then $\hat{M}_{\text{LRT}} - M_0 = o_p(1)$.

Assumptions 6(a)–(e) ensure the consistency and asymptotic normality of $\hat{\vartheta}_M$ for $M \leq M_0$, where (c)–(e) correspond to Assumption A6 of White (1982). Per Assumption 6(f), the Kullback–Leibler information criterion of the model relative to the true M_0 -component model strictly decreases as the number of components M increases for $M < M_0$.

Selecting the significance level q_n in finite samples is challenging. Due to the complexity of analyzing the optimal choice of q_n , we leave the formal analysis for future research. The simulation results in Section 6 suggest that $q_n = 0.01$ provides good finite-sample performance; we also recommend presenting results across conventional levels (1%, 5%, 10%).

To compare BIC and LRT performance across different q_n values, we also recommend conducting a simulation exercise similar to a parametric bootstrap; researchers can gener-

ate datasets from the estimated model with M components, apply methods with different values of q_n , and analyze the frequency distribution of estimated components as shown in Tables 1–B3 in Section 6. We show the inconsistency of AIC and consistency of BIC in Appendix A.9.

5.4 A Nonparametric Alternative: Rank-Based Lower Bound

The methods in Sections 5.1–5.3 rely on parametric specifications of the component densities $f(w; \theta_j)$. As a robustness check—particularly useful when the parametric assumption is in doubt—we briefly outline a nonparametric approach that estimates a *lower bound* for M_0 without imposing distributional assumptions.

The rank test method. Following Kasahara and Shimotsu (2014), we partition the support of each Y_t into mutually exclusive subsets $\Delta_t = \{\delta_1^t, \dots, \delta_{|\Delta_t|}^t\}$ and construct bivariate probability matrices $\mathbf{P}_k$ (equation 100) whose rank identifies the lower bound of M under the conditional independence structure of model (1). Specifically, $\mathbf{P}_k = \sum_{j=1}^M \alpha^j \mathbf{p}_k^j (\mathbf{q}_k^j)^\top$, implying $\text{rank}(\mathbf{P}_k) \leq M$, with equality under suitable rank conditions on the component-specific probability vectors.

Test statistic and implementation. The rank-test statistic of Kleibergen and Paap (2006) (Proposition A.7) provides a χ^2 test for $\text{rank}(\mathbf{P}_k) \leq r$. When $T \geq 3$, we aggregate evidence across different choices of $k \in \{1, \dots, T\}$ using $\max\text{-rk}(r) = \max\{\text{rk}^1(r), \dots, \text{rk}^T(r)\}$ and $\text{ave-rk}(r)$ statistics. Bootstrap p-values are obtained via the Bayesian bootstrap, and a sequential testing procedure (see Kasahara and Shimotsu (2014) Section 3.2 and Appendix A.11) yields $\widehat{M}_{\text{rank}}$, a consistent estimator of a lower bound for M_0 .

This method complements our parametric LRT by providing a specification-free diagnostic. In the empirical application (Section 7), we report $\widehat{M}_{\text{rank}}$ alongside $\widehat{M}_{\text{LRT}}$ and $\widehat{M}_{\text{BIC}}$ to assess robustness.

6 Simulation

This section evaluates the performance of LRT, AIC, BIC, and rank tests (ave-rk, max-rk) in estimating the number of components. All estimation uses the default regularisation described in Section 5.2: a variance floor $\sigma_j^2 \geq 0.05 \hat{s}^2$, an α -penalty ($a_\alpha = 3$) on the mixing proportions α_j , and a τ -penalty ($a_\tau = 1.5$) on the sub-component weights τ_{jk} . Parameters are estimated by penalised MLE using the EM algorithm (Appendix A.8).

Appendix B.1 validates bootstrap LRT size and power under controlled DGPs (Table B1): LRT maintains near-nominal size (4.2–4.6%) and achieves superior power over rank tests across all scenarios tested.

We evaluate performance using DGPs with $M_0 = 3$ components, calibrated to Chilean manufacturing data at realistic sample sizes, examining robustness to error density misspecification. The choice of $M_0 = 3$ provides a common benchmark across industries; the DGP parameters are drawn from preliminary three-component estimates for each industry.

Table 1 reports under-selection ($\hat{M} < M_0$), correct selection ($\hat{M} = M_0$), and over-selection ($\hat{M} > M_0$) rates across 100 simulations for all three industries ($T = 3$). Panel A uses a normal-error DGP ($K_\epsilon = 1$); Panel B uses a mixture-error DGP ($K_\epsilon = 2$). Each row shows performance under both correctly-specified and misspecified error densities.

Table 1: Selection accuracy (%): under/correct/over-selection for $M_0 = 3$ ($T = 3$; Metal $n = 277$, Food $n = 1029$, Textiles $n = 235$)

Method	Panel A: $K_\epsilon = 1$ (Normal) DGP									Panel B: $K_\epsilon = 2$ (Mixture) DGP								
	Metal (381)			Food (311)			Textiles (321)			Metal (381)			Food (311)			Textiles (321)		
	<	=	>	<	=	>	<	=	>	<	=	>	<	=	>	<	=	>
<i>Normal error estimation ($K_\epsilon = 1$)</i>																		
AIC	0	79	21	0	79	21	0	91	9	0	1	99	0	0	100	0	24	76
BIC	0	100	0	0	100	0	0	100	0	0	24	76	0	0	100	0	78	22
LR ($q_n = 0.01$)	0	98	2	0	97	3	0	99	1	0	18	82	0	4	96	0	61	39
LR ($q_n = 0.05$)	0	90	10	0	95	5	0	94	6	0	12	88	0	0	100	0	42	58
LR ($q_n = 0.10$)	0	84	16	0	91	9	0	92	8	0	11	89	0	0	100	0	34	66
<i>Mixture error estimation ($K_\epsilon = 2$)</i>																		
AIC	0	81	19	0	78	22	0	82	18	0	76	24	0	31	69	0	69	31
BIC	0	100	0	0	100	0	0	100	0	0	98	2	0	95	5	0	100	0
LR ($q_n = 0.01$)	30	70	0	94	6	0	95	5	0	13	86	1	8	92	0	93	6	1
LR ($q_n = 0.05$)	10	84	6	3	93	4	31	64	5	1	97	2	0	100	0	55	36	9
LR ($q_n = 0.10$)	1	87	12	1	83	16	11	78	11	1	97	2	0	100	0	19	65	16
ave-rk (0.05)	69	31	0	93	7	0	96	4	0	49	51	0	93	7	0	96	4	0
max-rk (0.05)	80	20	0	95	5	0	96	4	0	66	34	0	93	7	0	96	4	0

Notes: 100 replications. “<”, “=”, “>”: percentage selecting $\hat{M} < M_0$, $\hat{M} = M_0$, $\hat{M} > M_0$ ($M_0 = 3$). DGP parameters calibrated to three-component estimates for each industry (available in the replication package). Full selection frequencies under the mixture-error DGP in Appendix Table B3.

Under the normal-error DGP (Panel A) with correctly-specified normal estimation, BIC and LR ($q_n = 0.01$) achieve near-perfect correct selection (97–100%). Over-specifying the error density (mixture estimation on a normal DGP) preserves BIC’s accuracy (100% correct across all industries) but causes LR ($q_n = 0.01$) to collapse into severe under-selection, with correct-selection rates dropping to 5–70% (Metal 70%, Food 6%, Textiles 5%). Under the mixture-error DGP (Panel B), misspecifying the error density as normal produces pervasive over-selection: BIC is correct in only 0–78% of replications (Metal 24%, Food 0%,

Textiles 78%) and LR ($q_n = 0.01$) over-selects in 39–96% of cases. Correctly-specified mixture estimation on the mixture DGP restores BIC robustness (95–100% correct), while LR ($q_n = 0.01$) is reliable for Metal and Food (86–92% correct) but under-selects for Textiles (only 6% correct). These patterns are confirmed by additional simulations under the mixture-error DGP (Appendix Table B3) and size tests validating the baseline regularisation (Appendix Table B5). Overall, BIC with mixture-error estimation provides the most reliable baseline across both DGPs; LR ($q_n = 0.01$) is sensitive to error-density misspecification in both directions, and LR ($q_n \in \{0.05, 0.10\}$) offers a more conservative complement.

7 Empirical Application: Chilean Manufacturing Plants

We apply the finite-mixture production-function framework to plant-level panel data from Chilean manufacturing, focusing on three industries where at least 150 plants are observed: Food Products (CIIU 311), Fabricated Metal Products (CIIU 381), and Textiles (CIIU 321). For each industry we construct two balanced panels: a short panel spanning 1992–1996 ($T=5$, used in Section B.4.4) and a long panel spanning 1987–1996 ($T=10$, the main analysis). A sequential bootstrap LRT selects the number of latent technology types $\hat{M}$; conditional on $\hat{M}$, per-type production elasticities are estimated via the Gandhi–Navarro–Rivers (GNR) two-step procedure (Gandhi et al., 2020). We find $\hat{M} = 2$ –6 latent technology types across the three industries at $T=10$ (2–7 across both panel lengths), with per-type materials elasticity intercepts spanning 0.44–0.74 — a range of heterogeneity that single-technology production-function estimators cannot recover by construction. Strikingly, the $\hat{\alpha}$ -weighted average of these per-type elasticities matches the pooled $M=1$ estimate in all three industries: conventional single-technology estimators recover the correct industry-level mean but conceal the underlying technology distribution.

7.1 Data and Descriptive Evidence

The data are drawn from Chile’s Annual National Industrial Survey (ENIA). All monetary variables are deflated to 1986 pesos. We use the same three industries as Gandhi et al. (2020) and Kasahara and Rodrigue (2008): Food Products (CIIU 311, a material-intensive sector), Fabricated Metal Products (CIIU 381, capital-intensive), and Textiles (CIIU 321).

Table 2 presents summary statistics for the $T=5$ balanced panel (1992–1996). The key observation is **substantial within-industry dispersion in the materials revenue share** $s_{it} = M_{it}/Y_{it}$, with standard deviations of 0.14–0.20 and interquartile ranges of 0.16–0.25 — large relative to means of 0.52–0.65. Figure 1 plots the firm-level time average $\bar{s}_i = (1/T) \sum_t s_{it}$ for the $T=10$ panel: averaging over years removes transitory shocks, so the remaining dispersion ($\sigma=0.16$ –0.29 across industries) reflects permanent differences

in material intensity across firms. Under a single homogeneous technology ($M=1$), this permanent variation would be absent. The observed spread motivates a model with multiple latent technology types that differ in their materials elasticity.

Table 2: Summary Statistics by Industry ($T=5$ panel, 1992–1996)

	Fabricated Metal (CIU 381)	Food Products (CIU 311)	Textiles (CIU 321)
<i>Panel A: Sample characteristics (1992–1996)</i>			
Observations	1,389	5,146	1,180
Plants (N)	279	1,030	237
Avg. years/plant	4.98	5.00	4.98
<i>Panel B: Materials revenue share $s_{it} = M_{it}/Y_{it}$</i>			
Mean (SD)	0.516 (0.186)	0.654 (0.143)	0.568 (0.198)
25th–75th pct	[0.389, 0.635]	[0.577, 0.741]	[0.439, 0.693]
<i>Panel C: Other variables (log levels; mean with SD)</i>			
$\ln Y_{it}$ (gross output)	13.29 (1.37)	13.01 (1.63)	13.35 (1.34)
$\ln K_{it}$ (capital)	11.55 (1.65)	11.00 (2.03)	11.86 (1.65)
$\ln L_{it}$ (labour cost)	11.49 (1.27)	10.83 (1.41)	11.30 (1.37)
$T=10$ panel: N	157	669	162

Notes: Chilean ENIA. Monetary variables deflated to 1986 pesos. Standard deviations in parentheses. Panel A covers all plants observed at least once in 1992–1996 (unbalanced sample; $N=279$, 1,030, and 237 for Metal, Food, and Textiles respectively). The balanced $T=5$ panel (all five years present), used for model selection in Appendix B.4.4, comprises $N=242$, 862, and 203 plants. Bottom row: number of plants in the long ($T=10$, 1987–1996) balanced panel.

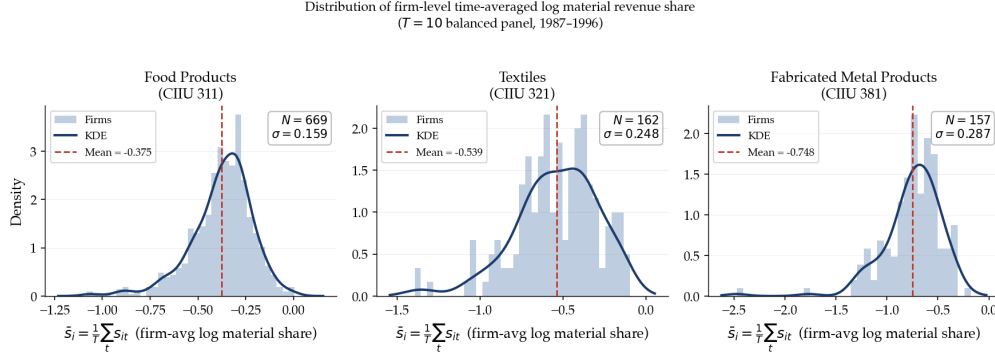


Figure 1: Distribution of firm-level time-averaged log materials revenue share $\bar{s}_i = \frac{1}{T} \sum_{t=1}^T s_{it}$, $s_{it} = \ln(M_{it}/Y_{it})$, across plants within each industry ($T=10$ balanced panel, 1987–1996; one observation per firm). Solid curve: kernel density estimate. The dispersed, non-Gaussian distributions reflect permanent unobserved heterogeneity across firms and motivate a finite-mixture model with multiple latent technology types.

Log inputs ($\tilde{k}_{it}, \tilde{\ell}_{it}, \tilde{m}_{it}$) are grand-mean centred within each industry–panel cell. Four-digit CIU sub-industry dummies enter as common (across-type) regressors in the GNR Step 1 share equation.

7.2 Baseline Specification

The **baseline specification** is the Nonlinear Mixture model with a two-component normal mixture residual ($K_\epsilon=2$, no Correlated Random Effects (CRE) augmentation). Under the GNR timing protocol (Gandhi et al., 2020), Hicks-neutral productivity ω_{it} cancels from the materials share first-order condition, so the share-equation system is free of endogeneity bias. Per-type coefficients ($\beta_{0,j}, \beta_{MM,j}, \beta_{LM,j}, \beta_{KM,j}$) are estimated by EM, yielding a linear materials elasticity surface $\bar{\epsilon}_{M,j}(\tilde{m}, \tilde{\ell}, \tilde{k}) = \beta_{0,j} + \beta_{MM,j}\tilde{m} + \beta_{LM,j}\tilde{\ell} + \beta_{KM,j}\tilde{k}$, which feeds into the GNR Step 2 (cubic-Markov GMM) to recover capital and labour elasticities. Robustness to the CRE (Mundlak) augmentation, Normal vs. Mixture errors, and winsorisation is reported in Appendix B.4.

7.3 Model Selection: Number of Latent Technology Types

Table 3 reports the selected number of types $\hat{M}$ from the sequential bootstrap LRT under the baseline Nonlinear Mixture ($K_\epsilon=2$) specification, alongside Cobb-Douglas and Linear Translog alternatives — each estimated under both Normal and Mixture error structures. The sequential test uses $B=199$ parametric bootstrap replicates at the 5% nominal level; the H_0 null uses unconstrained EM while H_1 applies a light Dirichlet regularisation

(Dirichlet-prior concentration $a=1.5$, τ -prior $b=1.1$, σ^2 floor $0.01\widehat{\text{Var}}(y)$) to prevent degenerate entropy-inflated components from raising the log-likelihood without sharpening the type partition.

Table 3: Sequential bootstrap LRT $\hat{M}$, baseline (no CRE). Three functional forms $\times$ {Normal, Mixture} error structures; $B=199$, 5% level. Bold: preferred Mixture specification at each flexibility level.

Panel A: $T = 10$ (main results, $N_{\text{Food}} = 669$, $N_{\text{Metal}} = 157$, $N_{\text{Tex}} = 162$)

Spec (per-type q)	Error dist.	Metal	Food	Textiles
CD ($q=0$)	Normal	9	7	9
CD ($q=0$)	Mixture	3	3	4
Linear ($q=3$)	Normal	6	2	6
Linear ($q=3$)	Mixture	4	4	5
Nonlinear ($q=3$)	Normal	4	7	2
Nonlinear ($q=3$)	Mixture	2	6	2

Panel B: $T = 5$ robustness (see Appendix B.4.4; $N_{\text{Food}} = 862$, $N_{\text{Metal}} = 242$, $N_{\text{Tex}} = 203$)

Spec	Error dist.	Metal	Food	Textiles
Nonlinear ($q=3$)	Mixture	2	7	3

Notes. All specifications include 4-digit CIIU sub-industry dummies as common (across-type) regressors. Nonlinear uses the inside-log design with a Linear-Mixture warm start. The light-penalty H_1 avoids degenerate solutions; H_0 is always unpenalised. Raw (no-winsor) data. Less flexible functional forms (CD, Linear) push selectors toward larger $\hat{M}$ because they absorb type-heterogeneous curvature via extra components. Panel B shows no-CRE $T=5$ robustness. The bolded Nonlinear Mixture rows of Panel A (Metal: 2, Food: 6, Textiles: 2) serve as the Phase 2 cluster-bootstrap anchors for Figures 2–3. Winsorisation and CRE margins are reported in Appendix B.4.2 and B.4.1.

Three patterns emerge. First, the Nonlinear Mixture model selects $\hat{M}=2$ in Metal and Textiles, and $\hat{M}=6$ in Food at $T=10$ — consistent under the raw (no-winsor) data. Second, less flexible functional forms tend to select larger $\hat{M}$: at $T=10$, Normal Cobb-Douglas selects 7–9, suggesting that type-heterogeneous curvature is being absorbed by extra mixture components when the mean function is too rigid. Third, under-selection ($\hat{M}<2$) is negligible across all Mixture specifications at $T=10$, consistent with meaningful latent technology heterogeneity in all three industries.

The selected $\hat{M}$ is robust to the error distribution: Table 3 shows that Cobb-Douglas and Linear Translog specifications select comparable $\hat{M}$ under Mixture errors, while Normal errors over-select by one to several types (Appendix B.4.3). Input winsorisation and CRE augmentation are robustness margins documented in Appendices B.4.2 and B.4.1. The Nonlinear Mixture baseline is therefore the preferred anchor for the GNR second stage.

7.4 Latent Technology Types: Materials Elasticity Surface

A central finding of this section is that the $\hat{\alpha}$ -weighted aggregate materials elasticity $\varepsilon_M^{\text{agg}} = \sum_j \hat{\alpha}_j \varepsilon_{M,j}$ lies within the pooled $M=1$ GNR 95% confidence interval in all three industries (Table 4): the mixture model does not shift the industry-level mean but decomposes it into constituent technologies. The novelty lies in the per-type distribution characterised by the EM-estimated intercepts $\beta_{0,j}$ — each type’s materials elasticity evaluated at the industry-mean input bundle ($\tilde{m}=\tilde{l}=\tilde{k}=0$, Appendix Table B13). Figure 2 plots $\beta_{0,j}$ for all types in each industry with conditional 95% Phase 2 cluster-bootstrap confidence intervals (CIs conditioned on the LRT re-selecting $\hat{M}$).⁷

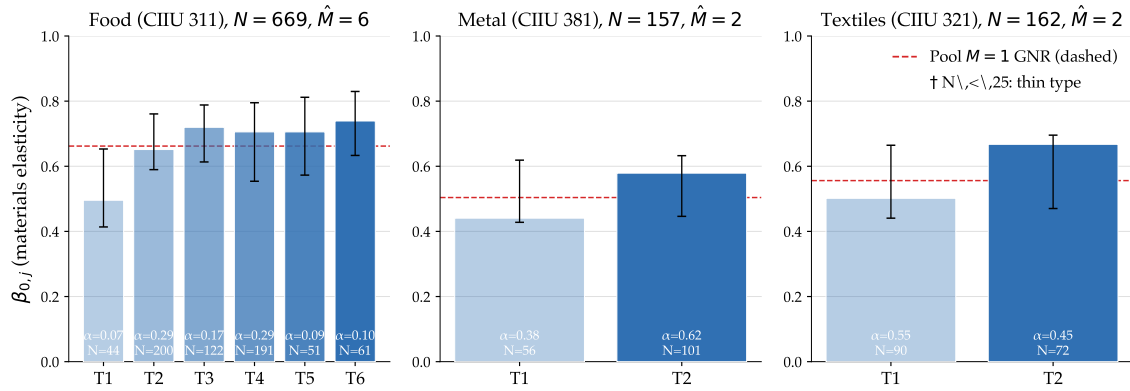


Figure 2: Per-type materials intercept $\beta_{0,j}$ (GNR Translog intercept = ε_M at centred inputs $\tilde{m}=\tilde{l}=\tilde{k}=0$), with conditional 95% cluster-bootstrap CIs (whiskers). Bars show data point estimates; types ranked left-to-right by $\beta_{0,j}$ (ascending); estimated weight $\hat{\alpha}_j$ shown inside each bar. Dashed horizontal line: pooled $M=1$ Python GNR estimate. Baseline: Nonlinear Mixture, $K_e=2$, no CRE, $T=10$. Types labelled † ($N < 25$) are thin types; interpret their CIs with caution.

Three patterns are visible. First, **heterogeneity is concentrated in the intercept** $\beta_{0,j}$: across types within a cell, $\beta_{0,j}$ varies by 0.11–0.24 (a 22–43% range around the pooled mean), while the input-interaction slopes ($\beta_{MM}, \beta_{LM}, \beta_{KM}$) vary by at most 0.13 and often overlap across types (see Appendix Table B13). Latent technology types are identified by their *level* of materials intensity, not by differential curvature in input substitution. Second, conditional CI widths on $\beta_{0,j}$ are available for all types with $N_j \geq 25$, confirming that the type partition is statistically sharp for well-populated types. Third, $\beta_{MM} < 0$ everywhere (diminishing returns to materials within a type) and $\beta_{KM} > 0$ everywhere

⁷Phase 2 cluster-bootstrap uses $B=350$ outer firm-level replicates; L2-Hungarian matching assigns type labels across replicates. Full per-type Step 1 coefficient tables are in Appendix B.4.5.

(capital–materials complementarity), confirming the monotonicity properties of the GNR estimator.

7.5 Production Elasticities and Returns to Scale

Figure 3 plots the full set of output elasticities ($\bar{\epsilon}_K, \bar{\epsilon}_L, \bar{\epsilon}_M$) and returns to scale ($\widehat{RTS}$) across types, with conditional 95% cluster-bootstrap CIs. The dashed lines mark the pooled $M=1$ GNR point estimates.

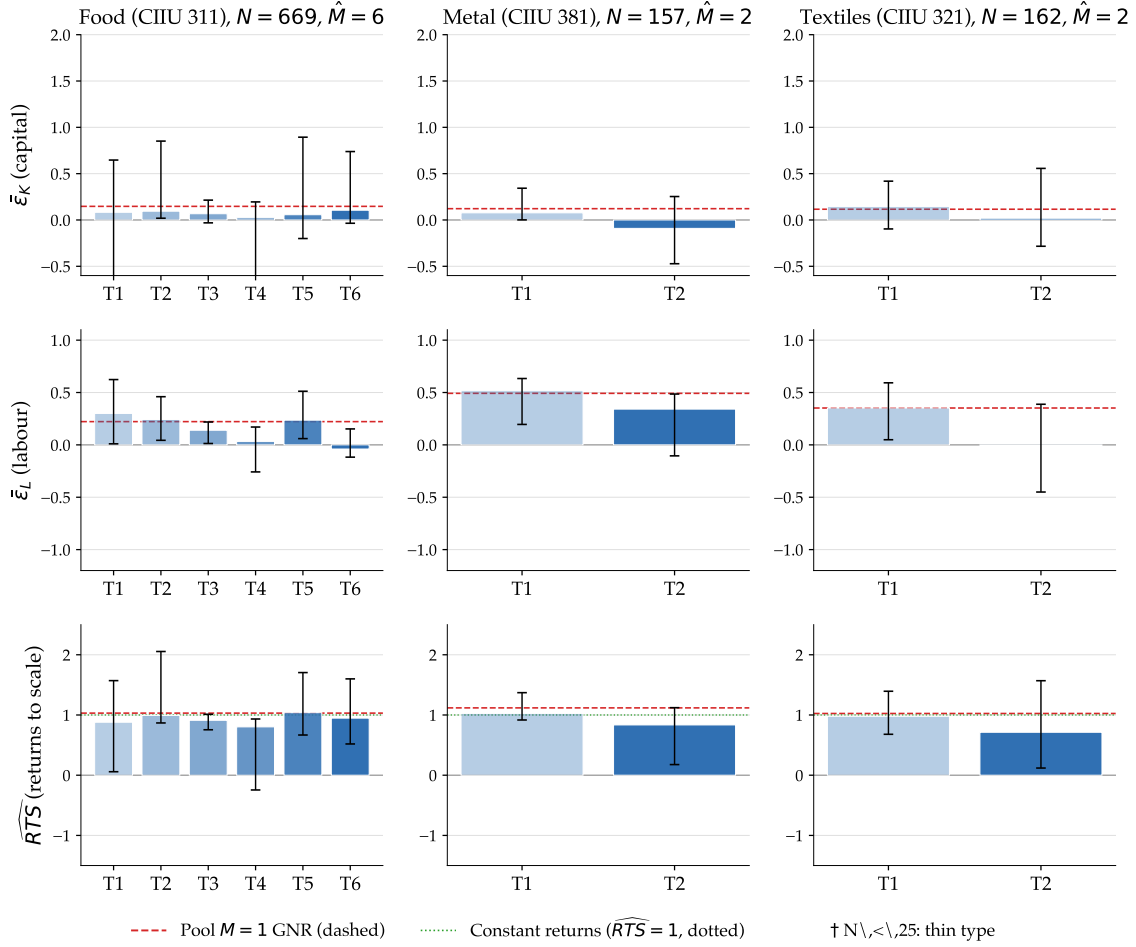


Figure 3: Per-type GNR output elasticities and returns to scale ($T=10$, Nonlinear Mixture ($K_\epsilon=2$), no CRE). Each column is one industry; rows show $\bar{\epsilon}_K$, $\bar{\epsilon}_L$, and $\widehat{RTS}$. Bars: data point estimates; whiskers are conditional 95% cluster-bootstrap CIs. Red dashed line: pooled $M=1$ Python GNR estimate. Green dotted line in bottom row: constant returns ($\widehat{RTS}=1$). Types labelled † have $N < 25$ plants (interpret their CIs with caution). Elasticities evaluated at the industry-wide grand mean. The GNR monotonicity projection enforces $\epsilon_K \geq 0$ at every plant-year observation, so per-plant capital elasticities are non-negative by construction.

Two features of Figure 3 warrant comment. First, capital elasticities are imprecisely estimated across most types: nine of ten type-level unconditional 95% CIs include zero, with Metal T2 ($\hat{\alpha}=0.62$, $N=101$) recording a slightly negative point estimate ($\bar{\epsilon}_K = -0.09$, CI: $[-0.54, +0.31]$). This does not violate the GNR monotonicity constraint: the SLSQP

projection enforces $\varepsilon_K \geq 0$ at every plant-year observation at the centred input bundle; a slightly negative type-level average can arise when the dominant type's mean input mix sits near the constrained boundary and bootstrap resampling shifts type assignments. Capital plays a secondary role in technology-type differentiation; types separate primarily along materials intensity $\beta_{0,j}$. Second, Food T1 is the thinnest type ($\hat{\alpha}=0.07$, $N=44$) and carries correspondingly wider bootstrap confidence intervals, consistent with the general pattern that minority types carry greater estimation uncertainty.

Table 4 quantifies the pooled benchmarks and the $\hat{\alpha}$ -weighted aggregate $\bar{\varepsilon}_k^{\text{agg}} = \sum_j \hat{\alpha}_j \bar{\varepsilon}_{k,j}$ with both conditional and unconditional Phase 2 cluster-bootstrap CIs.

Table 4: GNR elasticities at the sample mean ($T=10$): pooled benchmarks and $\hat{\alpha}$ -weighted mixture aggregate. Pool 95% CIs: firm-level cluster-bootstrap ($B=500$ Python, $B=200$ gnrprod). Aggregate 95% CIs: Phase 2 v8 no-CRE unconditional cluster-bootstrap (all B replicates). † CI contains zero.

Source	$\bar{\varepsilon}_K$	$\bar{\varepsilon}_L$	$\bar{\varepsilon}_M$	$\widehat{\text{RTS}}$
Food CIU 311, $T=10, N=669, \hat{M}=6$				
Pool $M=1$, Python GNR	+0.147	+0.222	+0.662	+1.031
95% CI	[+0.089, +0.281]	[+0.186, +0.591]	[+0.654, +0.671]	[+0.952, +1.488]
Pool $M=1$, gnrprod R	+0.105	+0.256	+0.662	+1.023
95% CI	[+0.090, +0.123]	[+0.219, +0.287]	[+0.654, +0.670]	—
Agg. $\sum_j \hat{\alpha}_j$ (data)	+0.066	+0.140	+0.709	+0.916
95% CI ($N_b=350$)	[−0.090, +0.243]†	[+0.038, +0.202]	[+0.692, +0.739]	[+0.706, +1.114]
Metal CIU 381, $T=10, N=157, \hat{M}=2$				
Pool $M=1$, Python GNR	+0.122	+0.492	+0.504	+1.118
95% CI	[−0.092, +0.278]†	[+0.198, +0.733]	[+0.483, +0.528]	[+0.669, +1.415]
Pool $M=1$, gnrprod R	+0.129	+0.500	+0.504	+1.133
95% CI	[+0.084, +0.175]	[+0.393, +0.563]	[+0.483, +0.528]	—
Agg. $\sum_j \hat{\alpha}_j$ (data)	−0.027	+0.408	+0.526	+0.908
95% CI ($N_b=900$)	[−0.230, +0.382]†	[+0.131, +0.565]	[+0.499, +0.589]	[+0.524, +1.267]
Textiles CIU 321, $T=10, N=162, \hat{M}=2$				
Pool $M=1$, Python GNR	+0.116	+0.352	+0.556	+1.025
95% CI	[+0.054, +0.172]	[+0.179, +0.473]	[+0.535, +0.580]	[+0.841, +1.173]
Pool $M=1$, gnrprod R	+0.114	+0.374	+0.556	+1.044
95% CI	[+0.075, +0.142]	[+0.304, +0.431]	[+0.536, +0.579]	—
Agg. $\sum_j \hat{\alpha}_j$ (data)	+0.089	+0.193	+0.578	+0.860
95% CI ($N_b=350$)	[−0.121, +0.411]†	[−0.128, +0.435]†	[+0.522, +0.612]	[+0.461, +1.286]

Notes. Elasticities at the grand-mean-centred input mean ($\tilde{m}=\tilde{\ell}=\tilde{k}=0$). Raw (no-winsor) data. Pool CIs from standalone cluster-bootstraps: Python 11-param GNR polynomial (gnr_per_type_exact, $B=500$) and gnrprod R (Jin, 2023) ($B=200$); gnrprod R does not output a joint RTS CI (—). Aggregate: $\hat{\alpha}$ -weighted mean; CIs from Phase 2 no-CRE bootstrap (firm-level resampling, $B=350$, $B_{\text{in}}=19$ inner LRT, L2-Hungarian type relabelling; Metal $B=900$, Food/Textiles $B=350$). † CI contains zero.

7.6 Discussion

Materials elasticity. The pooled $M=1$ materials elasticity lies at 0.50–0.66 across industries at $T=10$ (Table 4), consistent with the GNR Chilean and Colombian benchmarks (Gandhi et al., 2020). The per-type distribution in Figure 2 reveals meaningful heterogeneity: in Food ($\hat{M}=6$), six types span a wide range in $\beta_{0,j}$ from 0.50 to 0.74; in Metal ($\hat{M}=2$)

and Textiles ($\hat{M}=2$), two types each show a clear low-high contrast in materials intensity (Metal: $\beta_{0,j} \in \{0.441, 0.579\}$; Textiles: $\{0.502, 0.667\}$). The $\hat{\alpha}$ -weighted aggregate $\bar{\varepsilon}_M^{\text{agg}}$ lies within the pooled $M=1$ GNR 95% confidence interval in all three industries (Table 4), as expected from the consistency of both estimators. The novelty of the mixture model lies not in moving the aggregate but in revealing that the aggregate conceals substantial within-industry heterogeneity in materials intensity that the pooled $M=1$ estimate cannot recover.

Capital and labour elasticities. The pooled $M=1$ capital elasticity excludes zero in four of six cells (Table 4); the exceptions are Metal $T=10$ (Python GNR CI $[-0.092, +0.278]$) and Textiles $T=5$, reflecting finite-sample imprecision in the cubic-Markov GMM step at small N or short T . The $\hat{\alpha}$ -weighted aggregate capital elasticity is small and statistically indistinguishable from zero in all three industries: $+0.07$ for Food, -0.03 for Metal, and $+0.09$ for Textiles, all with unconditional 95% CIs containing zero (Table 4). The monotonicity projection constrains $\varepsilon_K \geq 0$ at every plant-year observation; aggregate values at the grand mean reflect each type’s input mix relative to the industry centre. Across all plant-year observations, the SLSQP projection reduces the fraction with any violated monotonicity condition ($\bar{\varepsilon}_K < 0$, $\bar{\varepsilon}_L < 0$, or $\bar{\varepsilon}_M < 0$) from up to 21% (unconstrained) to 0.0% (constrained) for every type and industry, confirming that the production function satisfies the basic input-monotonicity requirements of production theory at the individual plant level. Labour elasticity, by contrast, excludes zero in all six pooled cells and in almost all per-type conditional CIs.

Returns to scale. The pooled RTS lies in $[0.875, 1.133]$ across all six industry-implementation pairs, consistent with approximately constant returns. The per-type RTS distribution (Figure 3, bottom row) shows wider dispersion in point estimates but CIs for thin types are correspondingly wide; the majority types (weight ≥ 0.25) consistently have RTS conditional CIs that include 1.0. The $\hat{\alpha}$ -weighted aggregate RTS lies in $[0.86, 0.92]$ across the three industries (Table 4), broadly consistent with constant returns; all three unconditional CIs overlap 1.0.

Productivity persistence. The Step 2 AR(1) persistence parameter $\hat{\rho}$ ranges from 0.47 to 0.96 across types at $T=10$, with conditional CIs that often reach 0.96–1.00 on the right. This near-random-walk productivity is consistent with the broader literature (Gandhi et al., 2020) and is not a failure of AR(1) identification. Type-level variation in $\hat{\rho}$ (not shown in the figures) reflects genuine heterogeneity in the speed at which idiosyncratic productivity shocks dissipate across latent technology groups.

Mixture vs. pooled benchmark. The primary empirical contribution of the mixture model is the revelation of within-industry heterogeneity that the single-technology estimator cannot recover by construction. In Food ($\hat{M}=6$), six types reveal substantial dispersion in materials intensity [$\beta_{0,j}$ values to be updated from Phase 2 v8 bootstrap]. In Metal and Textiles ($\hat{M}=2$ each), the low-vs-high type split carries a clear economic interpretation: one technology group is significantly more materials-intensive than the other at the same (grand-mean) input bundle. The Python GNR and `gnrprod` R pool estimates agree closely on all four elasticities, confirming that the pooled $M=1$ result is not an artefact of the implementation.

8 Conclusion

This paper develops likelihood-based methods for selecting the number of components in panel data finite mixture regression models. We show that panel data eliminates the higher-order degeneracy of cross-sectional normal mixtures while unbounded likelihood and infinite Fisher information persist. We derive the asymptotic null distribution of the LRT statistic, establish BIC consistency and AIC inconsistency, and propose a sequential testing procedure for consistent component selection.

Applying these methods to Chilean manufacturing plants, the sequential bootstrap LRT under the Nonlinear Mixture specification ($K_\epsilon=2$) selects $\hat{M}=2-6$ latent technology types per industry at $T=10$ (Table 3). Conditional on the selected partition, per-type GNR estimates reveal that the $\hat{\alpha}$ -weighted aggregate materials elasticity lies within the pooled $M=1$ GNR confidence interval in all three industries, while the per-type distribution reveals meaningful within-industry heterogeneity invisible to single-technology production-function estimators by construction. Type heterogeneity is concentrated in the level of materials intensity at the sample mean (the intercept $\beta_{0,j}$), not in curvature or input-substitution slopes — a pattern visible only after conditioning on the mixture partition. Aggregate returns to scale are close to constant in all industries, consistent with competitive output markets, while type-level estimates reveal meaningful heterogeneity in production scale properties.

Proofs and additional simulation and empirical results are provided in the Online Appendix.

Data Availability Statement

The Chilean Annual National Industrial Survey (ENIA) data used in this study are publicly available from the Instituto Nacional de Estadísticas (INE), Chile. Anonymized plant-level panels are included in the replication package deposited at Zenodo (DOI to be assigned).

upon acceptance). All code necessary to reproduce the results is included in the replication package.

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